

Techno-economic analysis of lithium-ion battery price reduction considering carbon footprint based on life cycle assessment

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ABSTRACT

Lithium-ion batteries (LIBs) play a crucial role in driving energy transitions, particularly in electric vehicles (EVs) and energy storage systems. Forecasting LIB prices has received significant attention due to the tightening of raw material markets. Additionally, the implementation of carbon pricing policies has highlighted the need to incorporate the carbon footprint of LIBs in price evaluation. This study presents a comprehensive analysis of the carbon footprint, pricing trajectories, and forthcoming carbon pricing ramifications associated with LIBs. Firstly, the findings derived from the life-cycle carbon footprint assessment indicate that the global production volume-weighted average carbon footprint of EV battery packs spans 49.2–81.1 kgCO₂eq/kWh, depending on the chosen end-of-life treatment. Strategic end-of-life selections, aligned with battery chemical composition, can diminish the carbon footprint by up to 46% relative to improper disposal methods. Notably, direct cathode recycling of NCA and NMC622 batteries exhibits superior environmental performance. Secondly, techno-economic analysis predicts that the mean price of EV battery packs with diverse chemical compositions will decline to \$75.1/kWh by 2030, factoring in the compound annual growth rate of critical raw material prices over the past decade. LFP batteries emerge as the top economic performers. Lastly, the results reveal that monetizing a battery's carbon footprint may offset battery price reductions through learning effects. This underscores the imperative of effective recycling for curtailing the carbon footprint and ensuring the enduring economic and environmental competitiveness of LIBs.

1. Introduction

The growing global awareness of climate change, attributed mainly to the combustion of fossil fuels, is accelerating the development of low-carbon energy technologies. The transition to renewable energy sources is vital to keeping the global temperature rise below 2 °C, and ideally, even below 1.5 °C, compared to pre-industrial levels. However, the inherent intermittency of renewable energy sources has resulted in the need for the integration of energy storage devices (Yekini Suberu et al., 2014). Among all energy storage technologies available, lithium-ion batteries (LIBs) have gained significant attention in various applications due to their long cycle life and high efficiency (Ritchie, 2021). These batteries find applications in a wide range of devices, from small-scale portable electronics to high-capacity electric vehicles (EVs) and stationary energy storage systems (SESS). Nevertheless, LIBs face several challenges in enabling the widespread electrification of the transport sector and providing stationary grid storage. The primary

hurdle is the high price of batteries, which hinders the broader acceptance of the critical medium for clean energy (Ritchie, 2021). In order to achieve price parity with conventional internal combustion engine vehicles in the absence of incentives, it is believed that the cost of EV battery packs should be less than \$80/kWh (US dollars per kilowatt-hour) (Vehicle Technologies Office, 2022). Similarly, for SESS, the European Strategic Energy Technology Plan has set a system-level price target of €150/kWh (Euros per kilowatt-hour) for achieving cleaner energy systems (Tsiropoulos and Tarvydas, 2018). Achieving affordable battery prices is pivotal in enabling energy transitions, such as their integration with photovoltaic systems (Ershad et al., 2021). While battery prices have experienced significant declines over the past decade, a critical question looms regarding the pace at which they will reach these targets, as this will profoundly shape the future landscape of transport modes and energy infrastructures. Recent data underscores this concern, indicating an increase in the price of EV battery packs from \$138/kWh in 2021 to \$151/kWh in 2022, attributed to surging raw material costs (BloombergNEF, 2022).

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Abbreviation and nomenclature

BOMs	bill of materials	HMR	hydrometallurgical recycling
C2P	cell to pack	LCA	life cycle assessment
C2S	cell to system	LFP	lithium iron phosphate
CAGR	compound annual growth rate	LIB	lithium-ion battery
CIPK	carbon intensity per kilowatt-hour	M2C	mineral to cell
DCR	direct cathode recycling	NCA	lithium nickel cobalt aluminum oxide
EB	electric bus	NMC	lithium nickel manganese cobalt oxide
ED	energy-designed	PD	power-designed
EoL	end-of-life	PMR	pyrometallurgical recycling
EV	electric vehicle	RR	recovery rate
GHG	greenhouse gas	SESS	stationary energy storage system
		TEA	techno-economic analysis

As of today, several researchers have developed learning curve-based models for battery price (or cost) projections. This techno-economic analysis method is widely embraced and of paramount importance for assessing the economic feasibility of energy technologies. In contrast to bottom-up models, learning curves require fewer input parameters and data, straightforward and pragmatic choice for capturing the trends in battery price reduction (Wentker et al., 2019; Greenwood et al., 2021). Nykvist and Nilsson (2015) analyzed 85 estimates reported between 2007 and 2014 to track the costs of LIBs at the pack level. They discovered that the learning rate, which represents the cost reduction when cumulative battery capacity doubles, ranged from 6% to 9%. In subsequent years, the same authors collected 25 new data points from 2015 to 2017 and updated the learning rate for the entire industry to $16 \pm 3\%$ (Nykvist et al., 2019). Schmidt et al. (2017) formulated experience curves to forecast future prices of 11 electrical energy storage technologies. Their findings indicated that the experience rates for LIBs were $30 \pm 3\%$ for electronics, $16 \pm 4\%$ for EVs, $12 \pm 4\%$ for residential-scale SESS, and $12 \pm 3\%$ for utility-scale SESS. Kittner et al. (2017) developed a two-factor model based on cumulative LIB capacities and technology innovation. They reported a learning rate of 16.9% for increasing production scale and an additional 2.0% reduction in LIB pack-level prices per 100 patent activities. Hsieh et al. (2019) introduced a two-stage learning curve model eyeing potential illogical price reductions, thus setting battery raw material costs as the learning floor for battery packs. They observed learning rates of 3.52% for active material synthesis and $16.49 \pm 4.52\%$ for battery pack production of lithium nickel manganese cobalt oxide (NMC) battery packs used in EVs. Penisa et al. (2020) accessed the learning curve models by examining multiple factors, such as cumulative battery capacity, patent activity, lithium metal price, and cobalt price. The most statistically robust outcomes emerged from a two-factor model with learning rates of 21.18% for battery demand and 3.0% for innovation.

Another challenge facing the widespread deployment of LIBs pertains to their environmental impact, as their manufacturing processes deplete resources and energy. The amount of carbon dioxide emitted during battery production varies depending on factors such as the battery chemicals used, the origin of materials, and the energy sources employed. In order to systematically evaluate the environmental effects of different production pathways, many studies have been carried out on the life cycle assessment (LCA) of batteries. It evaluates the environmental performance, such as greenhouse gas (GHG) emissions or energy consumption related to each stage of the life of a product or a service (Muralikrishna et al., 2017). The scope of LCA for batteries varies depending on the defined system boundary. The cradle-to-gate analysis considers emissions associated with the upstream raw material extraction and the battery manufacturing process (Kim et al., 2016). In contrast, the cradle-to-grave analysis encompasses the entire battery manufacturing supply chain, the battery usage pattern, and end-of-life (EoL) treatment, such as burning, landfill, and recycling (Wang et al.,

2022). Comparative battery LCA plays a pivotal role in expediting the transition towards a cleaner energy mix and transportation, offering decision-makers insights into more environmentally friendly manufacturing processes. Regarding the application of batteries in vehicles, the environmental impacts of EVs compared to internal combustion engine vehicles have been broadly examined across the globe (Notter et al., 2010; Qiao et al., 2017a). Although EVs have zero emissions in operation, the installation of batteries leads to an external carbon footprint. The burdens are determined by various factors, such as battery chemistries (Majeau-Bettez et al., 2011; Ambrose and Kendall, 2016; Yin et al., 2019), solvents used in battery manufacturing (Zackrisson, 2010), energy requirements during manufacturing (Dai et al., 2019), energy sources for production (Ellingsen et al., 2013), and assembly locations (Dunn et al., 2014; Oliveira et al., 2015; Degen and Schütte, 2022). For stationary applications, substantial research has examined the environmental impacts of different battery production technologies (Hiremath et al., 2015; Stougie et al., 2019; Le Varlet et al., 2020; Carvalho et al., 2021). More recently, greater attention has been devoted to the use stage (Jenu et al., 2020) and EoL stage (Dunn et al., 2012; Mohr et al., 2020). This shift in focus is a response to the escalating global interest in embracing circular economy practices. Additionally, battery recycling has garnered considerable research attention (Chen et al., 2022, 2023). Notably, the cascading utilization of spent EV batteries in stationary applications has demonstrated environmental benefits by extending the battery lifespan (Wang et al., 2022; Ahmadi et al., 2015; Quan et al., 2022; Tao et al., 2021; Richa et al., 2017).

The existing LIB applications has mostly focused on either economic or environmental performance. However, the design of sustainable energy systems requires considering feasibility and affordability in both aspects. Some prior studies have made efforts to concurrently assess the carbon emissions and economics of batteries, particularly within specific applications like electric mobility or stationary grid solutions (Philippot et al., 2019; Baumann et al., 2017). Notably, Ciez and Whitacre (2019) made significant strides by employing attributional life cycle analysis and process-based cost models to analyze carbon emissions, energy consumption, and costs associated with the manufacturing and recycling of three distinct lithium-ion battery types. However, their research scope is confined to the cell level, with a primary focus on economic analyses related to recycling, such as the assessment of re-lithiation costs. This compartmentalization poses challenges when attempting comparisons across various battery chemistries and applications due to inherent disparities in system boundaries, data sources, and underlying assumptions. Furthermore, as many countries increasingly integrate carbon pricing into their climate change mitigation strategies (Black et al., 2022), there is a pressing need to re-evaluate battery pack prices while factoring in these external effects. This study endeavors to bridge the research gap by conducting a comprehensive examination of the carbon footprint and techno-economic dimensions of batteries. Importantly, the potential impact of varying carbon pricing levels on future battery prices

is scrutinized, thereby providing invaluable insights into the multifaceted pricing dynamics of LIBs from both environmental and economic perspectives.

2. Methodology and data

The methodologies employed in this study are illustrated in Fig. 1. The first part of the study involves conducting a life-cycle carbon footprint assessment of LIBs used in EVs and SESS applications, which assesses the carbon footprint and considers various EoL treatments and battery chemistries. In the second part, a techno-economic analysis is conducted using a novel bootstrap-based two-stage learning curve model to project future battery prices at the pack level for EVs and the system level for SESS. The bootstrap method is employed to quantify the uncertainty associated with the learning rate regression; the two-stage model structure is designed to consider the practical constraints imposed by material costs in achieving battery price reductions. Subsequently, the carbon burdens are quantified in economic terms, enabling an exploration of the implications of carbon pricing on the future trend of LIB price reduction. This integration of environmental and techno-economic considerations sheds light on the implications of carbon pricing on the future costs of LIBs.

2.1. Carbon footprint assessment

Presently, the most widely recognized and credible method for calculating carbon footprints is the LCA approach (Gao et al., 2014). This approach is guided by the ISO 14040 and ISO 14044 standards, which provide a comprehensive framework and principles for implementing LCA (ISO 14040:2006, 2006; ISO 14044:2006, 2006). The LCA process consists of four standard phases aimed at evaluating the environmental performance of a product or service across its entire life cycle: goal and scope definition, inventory analysis, impact assessment, and interpretation.

2.1.1. Goal and scope definition

The primary objective of the carbon footprint assessment presented in this paper is to quantify the carbon footprint of LIBs across their complete life cycle. The specific emphasis is placed on evaluating the carbon footprint as the primary environmental impact category of interest. The functional unit used for this assessment is kilogram carbon dioxide equivalent per kilowatt-hour capacity ($\text{kgCO}_2\text{eq/kWh}$), encompassing the defined life cycle phases of LIBs. The system boundary of this study consists of most life cycle stages, including mineral acquisition, production of cells, assembling modules and packs, and the EoL phase.

However, it is crucial to note that two specific stages have been intentionally excluded from the analysis: transportation, due to limited standardized data availability across regions, and the use phase, where the battery undergoes charging and discharging reactions without direct emissions. Accounting for indirect emissions during the use phase presents a formidable challenge, as the carbon emission intensity of power supply systems can exhibit significant variation across different time periods and regions. Some regions prominently rely on renewable energy sources, while others are more reliant on fossil fuels. This variability complicates the development of a standardized framework for calculating these emissions. In addition, it's important to recognize that batteries, even those with the same chemical composition, can experience different levels of deterioration. This variance is heavily influenced by user behavior and preferences, impacting factors such as usage patterns and cycling frequency (Takeno et al., 2005). Given the intricate and multifaceted nature of these considerations, this study excludes indirect emissions associated within the battery's life cycle, avoiding potentially biased assumptions during the use phase.

2.1.2. Life cycle inventory (LCI) and life cycle impact assessment (LCIA)

Inventory analysis involves compiling and quantifying the energy and material flows associated with the different life cycle stages of a product or service (Qiao et al., 2017b; Fraval et al., 2019). For this study, the life cycle inventory data are acquired from multiple sources,

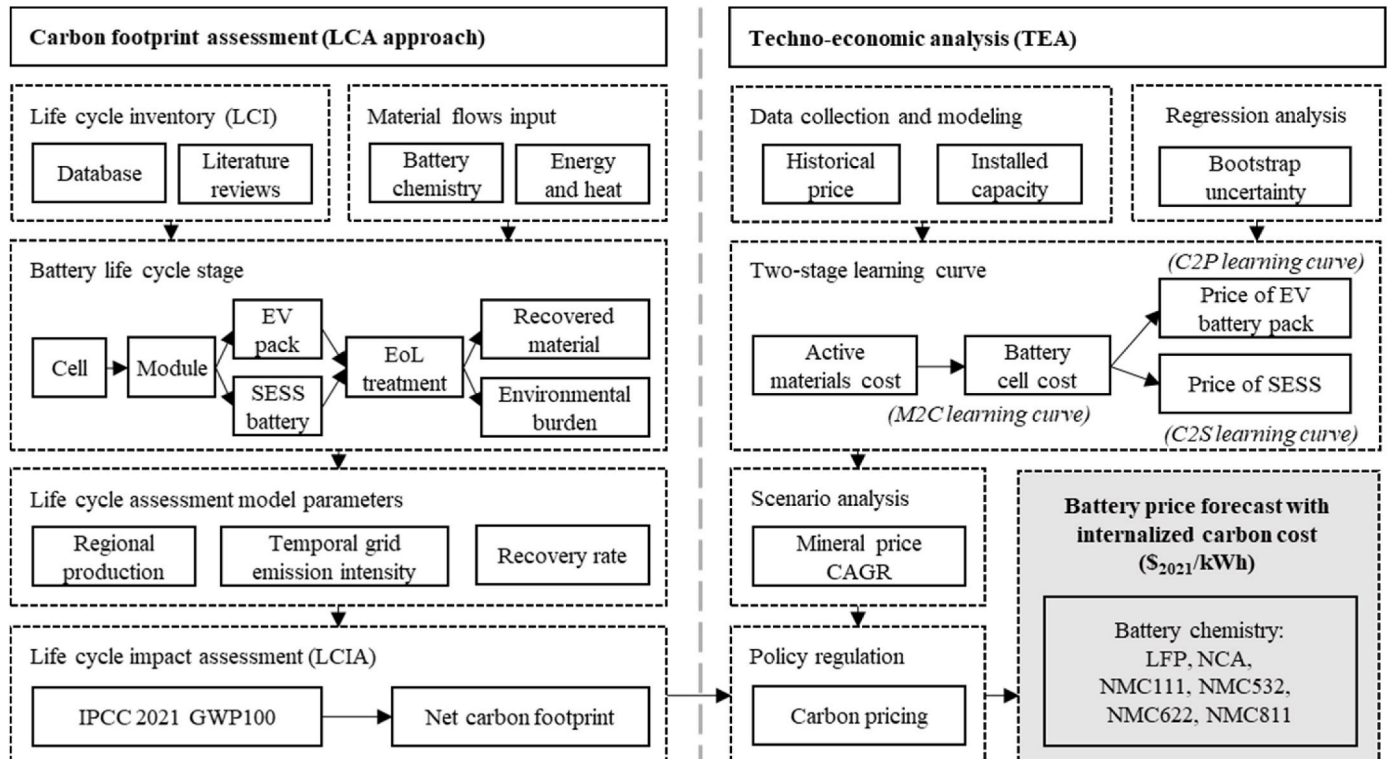


Fig. 1. Methodology framework of this study.

including the Ecoinvent 3.8 database (Wernet et al., 2016), Industry data 2.0 within SimaPro (Industry data LCA and library, 2022), and secondary data derived from existing literature (Quan et al., 2022; Tao et al., 2021; Yadav et al., 2021; Abbasi, 2014). The carbon footprint of LIBs is re-assessed by employing a script that utilizes the available data from the collected database. The Global Warming Potential (GWP100) provided by the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report Working Group (Arias et al., 2021) is adopted in the life cycle assessment. The GWP100 quantifies the heat absorbed by a unit mass of GHG in the atmosphere over a 100-year period (Environmental Protection Agency, 2016), resulting from emissions during the life cycle stage. This metric offers a classification of LCI results along with corresponding characterization factors. The product of these two elements can be used to quantify the extent of global warming potential. For example, 1 kg of methane from LCI results is converted and has a greenhouse gas effect equivalent to 29.8 kg of carbon dioxide. The characterization factors for other substances can be obtained from a recognized reference source (Cilleruelo, 2022). In order to facilitate effective comparison, the results are converted into carbon dioxide equivalents (CO₂eq), allowing for consistent evaluation across different GHGs.

2.1.3. Factors considered in carbon footprint assessment

To enable a comprehensive evaluation of the global carbon footprint of LIBs, this study considers a range of key factors, including battery chemistries, manufacturing regions, grid carbon intensity, battery applications, EoL treatments, and the material recovery rate in recycling. Firstly, regarding the composition of the battery cell, six representative cathode chemistries, namely LFP (lithium iron phosphate), NCA (lithium nickel cobalt aluminum oxide), and NMC (lithium nickel manganese cobalt oxide) of four kinds (NMC111, NMC532, NMC622, and NMC811, with numeric representations of the molar ratio) are investigated. Secondly, the study focuses on four major battery manufacturing regions—China (CN), European (EU), Japan and Korea (JK), and the United States (US) (Yu and Sumangil, 2022). These regions are selected to capture the geographical variations in raw material sources and electricity inputs during battery production. Furthermore, recognizing that each country or region is progressing towards a low-carbon society at different rates, it is crucial to account for the varying grid carbon intensities across different time periods for each manufacturing location. Grid carbon emission intensity refers to the amount of GHG emissions associated with producing 1 kWh of electricity. This study also explores a hypothetical scenario where all battery production processes utilize clean energy sources with zero carbon intensity per kilowatt-hour (0-CIPK). By making this assumption, the carbon emissions predominantly arise from the battery bill of materials (BOMs), providing insights into the potential reduction of carbon emissions in line with the ultimate decarbonization vision for power grids.

The assembly and design requirements of battery packs vary based on their specific battery applications, resulting in different outcomes. The scope of the carbon footprint assessment in this study focuses on evaluating batteries at the pack level, considering their applications in EVs for mobility and SESS in the grid or for energy storage purpose. Regarding the decommission of batteries, five EoL treatments are explored, namely incineration (burned and buried), landfill (direct disposal into a landfill without pre-treatment), direct cathode recycling (DCR; recovery of cathode materials directly from spent LIBs via electrolyte extraction), hydrometallurgical recycling (HMR; chemical recovery of critical metals through reactions in aqueous environments), and pyrometallurgical recycling (PMR; a smelting-like process with valuable metals recovered in the form of alloys) (Tao et al., 2021; Zhou et al., 2020). The first two treatments serve as benchmarks for improperly sorted and disposed waste batteries. In contrast, the latter three recycling processes are evaluated to assess their performance in recovering valuable materials as avoided products. This allows for an

assessment of the efficacy of each recovery method and its capacity to contribute to the circular economy. In addition, the degree of technological advancements in recycling plays a role in determining the successful reintegration of decommissioned batteries into the battery supply chain. This study investigates the carbon emission impacts of material recovery rates (RR) using baseline, high, and low scenarios for all three battery recycling methods. Detailed information on the data and assumptions used in the life cycle study, including life cycle inventories, grid carbon intensities, and recovery rates, can be found in [Supplementary information section 1](#).

2.2. Techno-economic analysis (TEA)

2.2.1. Price data collection and installed capacity estimation

The terms “price” and “cost” are often used interchangeably, as the price paid to upstream companies represents the cost incurred by the downstream supply chain. This study takes end-use as the central perspective; thus, the term “price” was consistently used at the level of pack or system. Other figures, including merchandising of minerals, battery cells, and the fee charged for the carbon footprint, are entitled to the “cost” for the purpose of differentiation. The battery prices serve as one of the key inputs in the regression analysis of learning curves. Those data are extensively gathered from online sources, including e-books, journal articles, publicly available government or commercial reports, related news, and websites. Through meticulous cross-referencing and eliminating redundant information, 320 data points as of 2021 are obtained at the level of battery cells, electric vehicle battery packs, and stationary energy storage systems. All price estimates have been adjusted for inflation to 2021 USD (\$) using the United States consumer price index (Bureau of Labor Statistics, 2022). The detailed data points and sources for prices can be found in the [Supplementary data](#) (which is an Excel file).

As the floor for the learning model, mineral costs are estimated based on historical trends. While critical mineral costs have exhibited a general upward trend in the past decade, they have also become increasingly volatile. In order to account for this unpredictability, this study adopts a base case scenario considering a ten-year compound annual growth rate (CAGR) for mineral costs. Additionally, it is crucial to consolidate historical battery usage and future installation demand as they serve as the driving force behind the learning curve analysis. Thus, this study collects existing market statistics, reviews relevant literature reports, and references policy planning targets to estimate the annual battery installed capacity up to 2030. [Supplementary information section 2](#) provides comprehensive details on the assumptions and data sources relevant to installation projections, chemistry share, and production share by region, which are crucial for conducting the techno-economic analysis.

2.2.2. Bootstrap-based two-stage learning curve model

The conventional one-stage learning curve model assumes that battery prices will exponentially decrease, eventually reaching zero. In order to capture the practical lower bounds imposed by material costs on battery prices, a two-stage learning curve model was developed by Hsieh et al. (2019) and serves as a major reference model for this paper. However, instead of utilizing active material costs as in the previous work, this study adopts battery cells as the intermediate factor bridging stage 1 and stage 2. The decision to use battery cells stems from the confidential nature of active material costs, which are not typically disclosed publicly and pose challenges for updating and expanding the learning curve model proposed by Hsieh et al. Accordingly, this study proposes that future prices of battery applications will converge toward battery cell costs while the battery cell costs themselves will approach a lower limit determined by mineral costs.

The proposed two-stage learning curve model is outlined in Equations. (1)–(6), utilizing cumulative battery installed capacities (V) as the driving factor for the battery price reductions. In the first stage of learning (M2C), battery cell costs (C) are determined by mineral costs

(M) and serve as a minimum threshold. These minerals comprise electrochemically active materials (i.e., anodes and cathodes), which vary depending on the battery chemistry type. In the second stage of learning, either from cell to EV pack (C2P) or SESS (C2S), the battery cell costs obtained from the first stage act as a price floor for LIBs utilized in electric mobility or stationary grid applications, respectively. One notable advantage of this model lies in its treatment of critical raw material costs as independent variables. This feature enhances its responsiveness, enabling it to promptly generate new battery price forecasts in response to evolving market dynamics.

2.2.2.1. Stage 1: from mineral to battery cell (M2C)

$$C_t = (C_0 - M_0) \times \left(\frac{V_{cell,t}}{V_{cell,0}} \right)^{b_{M2C}} + M_t \quad (1)$$

$$LR_{M2C} = 1 - 2^{b_{M2C}} \quad (2)$$

2.2.2.2. Stage 2: from battery cell to pack (C2P) or SESS (C2S)

$$P_{i,t} = (P_{i,0} - C_0) \times \left(\frac{V_{i,t}}{V_{i,0}} \right)^{b_{C2P,i}} + C_t \quad (3)$$

$$LR_{C2P,i} = 1 - 2^{b_{C2P,i}} \quad (4)$$

$$S_{j,t} = (S_{j,0} - C_0) \times \left(\frac{V_{j,t}}{V_{j,0}} \right)^{b_{C2S,j}} + C_t \quad (5)$$

$$LR_{C2S,j} = 1 - 2^{b_{C2S,j}} \quad (6)$$

where C_t : battery cell cost in year t .

M_t : anode and cathode active materials cost in year t .

V_t : cumulative battery installation capacities from the year 0 to the year t .

LR_k : learning rate for the learning stage k . It refers to the cost reduction followed by doubling cumulative production or capacity.

$P_{i,t}$: the price of battery packs in year t for electric mobility applications i , including electric vehicle (EV) and electric bus (EB)

$S_{j,t}$: the price of battery-based stationary energy storage systems¹ in year t for application design j , including energy-designed (ED) for utility-scale deployment and power-designed (PD) for commercial or residential-scale projects (Tsiropoulos and Tarvydas, 2018)

Due to the limited sample size ($N < 150$) and high uncertainty in battery prices for each application, the bootstrap method is utilized in the regression analysis. The bootstrapping technique involves randomly resampling data points from the original dataset with replacements to construct hypothetical populations, allowing us to quantify the uncertainty associated with the learning benefits for each specific use of LIBs. In this study, the bootstrap method is employed by repeating the sampling procedure 1000 times. The middle 95% of values are used to construct the confidence interval (CI) from the resulting bootstrap distribution. In order to provide clarity in the discussion, the median figure, which represents the value with the highest probability, is used consistently throughout the following chapters.

2.3. Carbon pricing

Carbon pricing is recognized as an effective strategy for addressing climate change by assigning a monetary cost to carbon emissions. This

policy approach helps shift the environmental burden to carbon producers, incentivizing the adoption of clean energy technologies and facilitating the transition to a low-carbon society. In response to growing environmental concerns and climate objectives, the European Union has introduced a carbon border adjustment mechanism (CBAM) that applies to selected imports from third countries (European Commission, 2022). The primary purpose of the CBAM is to prevent carbon leakage by harmonizing the carbon price between the European Union, which represents the largest carbon market globally, and countries outside the European Union. The implementation of the CBAM regulation underscores the significance of considering carbon footprint costs when projecting the future prices of LIBs. To account for the inherent uncertainty surrounding carbon pricing policies, an analysis of carbon price sensitivity is conducted, considering a range of carbon prices from \$0 to \$120 per kilogram of carbon dioxide equivalent (Stiglitz et al., 2017). This comprehensive range encompasses multiple scenarios and allows for a thorough exploration of the potential impacts of varying carbon prices.

3. Results and discussion

3.1. Life cycle assessment

3.1.1. Global average carbon footprint

Fig. 2 illustrates the global production volume-weighted averages of life cycle carbon emissions for LIBs used in EV applications in 2020, categorized by chemistry types and EoL treatments. The discussions focus on EV applications since the carbon footprint results for LIBs used in SESS applications are similar (see Fig. S4). The result shows that the carbon footprint for an EV battery pack is 81.1 kgCO₂eq/kWh with incineration, 71.9 kgCO₂eq/kWh with landfill, 49.2 kgCO₂eq/kWh with DCR, 65.9 kgCO₂eq/kWh with HMR, and 70.0 kgCO₂eq/kWh with PMR, based on the global market share of battery chemistry and production volume as weighting factors. These findings indicate that improper disposal of batteries leads to the highest carbon footprint for NMC111 due to the high embodied carbon in its BOMs, which exceeds that of LFP with landfill by 13.8% and NMC622 with incineration by 13.2%. Although the recycling process adds to the carbon footprint, the recovered secondary materials can be re-introduced into the battery supply chain, mitigating the need for virgin materials and resultant carbon emissions. Nevertheless, the extent of carbon footprint reduction varies substantially within different recycling methods and types of battery chemistry. Among the investigated recycling methods, DCR proves to be the most effective one. Compared to incineration, LIBs treated with DCR exhibit a reduction in life cycle carbon footprint ranging from 28% to 46%, depending on the battery chemistry. Conversely, PMR can result in net positive emissions due to intensive energy and chemical usage during recycling and is not suitable for LFP batteries due to the lack of valuable materials that can be recovered through this open-loop recycling process. Within the same EoL treatment, NMC batteries with a higher nickel content tend to have a lower carbon footprint, as they require less cobalt for production and recycling. However, as the nickel content increases (i.e., NMC811), carbon emissions rise due to substituting a higher carbon footprint lithium source used in precursor production (Tao et al., 2021). It can be traced back to the substitution of a lithium source with a higher carbon footprint during precursor production. When the lithium source is replaced from lithium carbonate to lithium hydroxide, the carbon emission of the lithium content increases by 360% (on a per kilogram cathode active material basis). Therefore, even though NMC811 uses less manganese and cobalt, its final carbon footprint is not reduced. Consequently, despite NMC811's reduced usage of manganese and cobalt, its final carbon footprint does not see a corresponding decrease (Table S4). This is mostly attributable to the assumptions of the referenced LCI source. This situation may evolve in the future as technological advancements and alternative lithium sources become available to battery manufacturers. As of now, NCA and

¹ In addition to battery, stationary energy storage systems (SESS) at the system level include the balance of system, energy management system, engineering, procurement and construction and power conversion system (Tsiropoulos and Tarvydas, 2018).

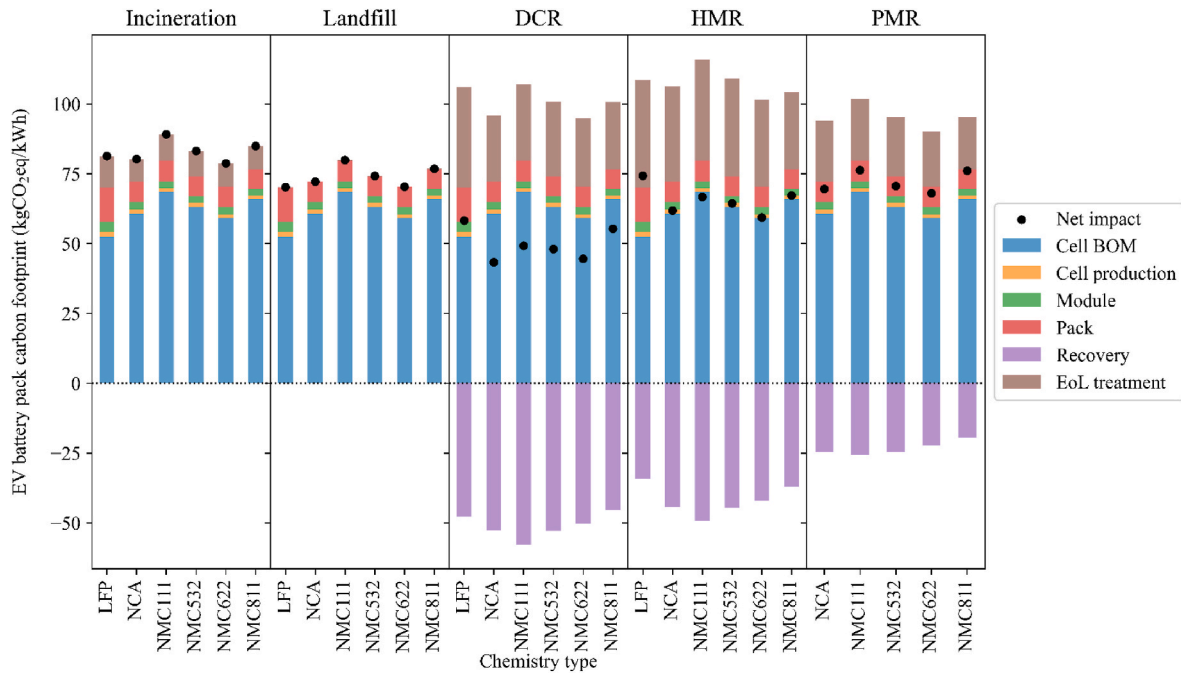


Fig. 2. Global average carbon footprint of the battery pack used in EVs in 2020 with five different EoL treatment methods: incineration, landfill, direct cathode recycling (DCR), hydrometallurgical recycling (HMR), and pyrometallurgical recycling (PMR).

NMC622 emerge as the two battery chemistries with lower carbon emissions compared to the others.

3.1.2. Carbon footprint variations

Fig. 3 illustrates the dynamic changes in the carbon emissions of EV battery packs over spatial and temporal variations in material and energy sources (see Fig. S5 for SESS results). The global average values in 2020 serve as a benchmark and are depicted by a solid black horizontal line. The spatial findings reveal that batteries in China have the highest

carbon emissions from a life cycle perspective. In contrast, it is envisaged that batteries in Europe have the best carbon performance over the life cycle stage until 2030 due to low grid carbon intensity. LIBs treated with the DCR and HMR methods demonstrate more significant carbon reduction potential when coupled with an improved grid carbon intensity, while incineration, landfill, and PMR possess limited improvement with clean energy sources. Global carbon emissions from batteries are projected to decrease by 7.1–10.6% from 2020 to 2030 (for each region: CN 5.6–9.2%, EU 4.5–7.6%, JK 1.7–2.9%, and US 8.6–14.5%)

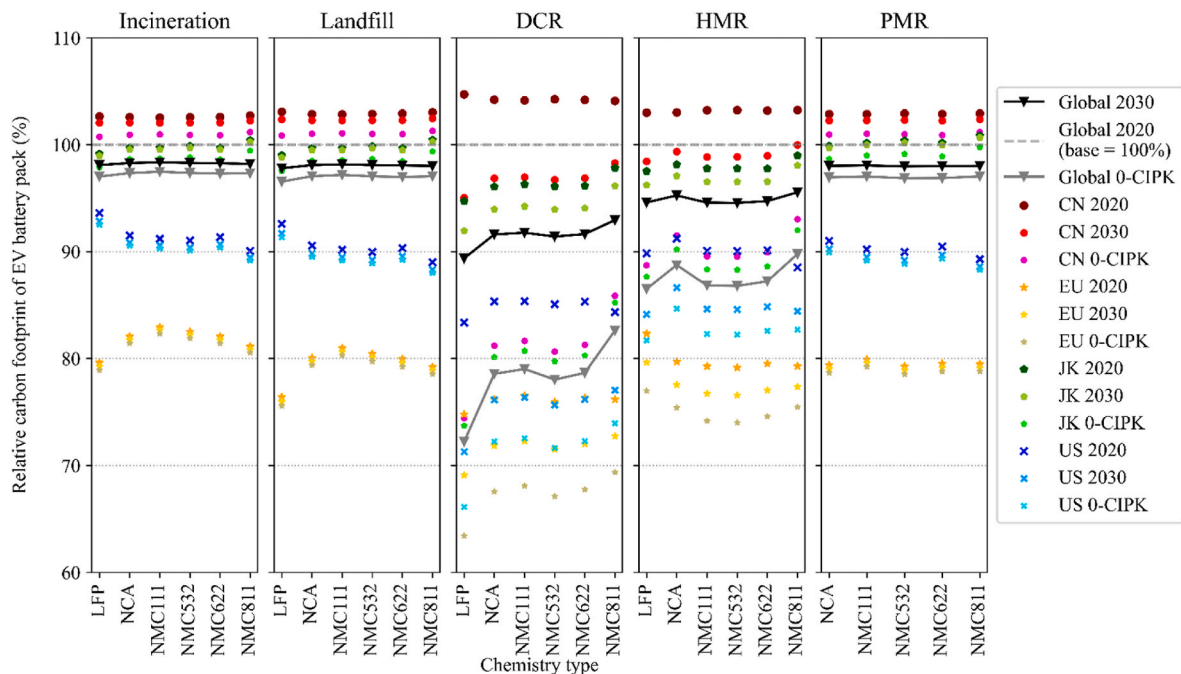


Fig. 3. Spatial and temporal variations in the carbon footprint of EV battery pack: Impacts of battery manufacturing regions and grid carbon emission intensity. Battery manufacturing regions include China (CN), European Union (EU), Japan & Korea (JK), and the United States (US). Temporal changes in grid carbon intensity include 2020, 2030, and zero carbon intensity per kilowatt-hour (0-CIPK).

through the utilization of the DCR recycling method with default recovery rate across battery types. Switching to clean energy throughout the battery supply chain (0-CIPK condition) can bring this downward trend to 17.4–27.8% (for each region: CN 17.5–28.9%, EU 9.0–15.2%, JK 12.9–22.2%, and US 12.3–20.7%). Variations in outcomes between regions indicate that the production location significantly influences the life cycle carbon footprint more than battery chemistry and EoL treatments. Consequently, adopting low-carbon battery BOMs emerges as another potential approach for reducing the carbon footprint of LIBs. Overall, the NCA battery's life cycle in Europe utilizing renewable clean energy and decommissioned via the DCR process demonstrates the lowest CO₂ emission value, followed by NMC622. Notably, as the carbon footprint of the raw material is subject to uncertainty in the decade, it is assumed that the carbon emissions from the battery's BOMs for each production region remain consistent, and the only variable that changed over time is grid carbon emission intensity.

The carbon footprint generated throughout the entire battery life cycle is also influenced by the advancements in recycling technology. Proper recycling of decommissioned batteries can help avoid the need for virgin materials, while the inadequate selection of recycling methods can lead to additional carbon emissions. As depicted in Fig. 3, considering a baseline recycling rate, all battery types show potential for carbon reduction, except for LFP batteries subjected to the HMR method. Fig. 4 presents the carbon footprints associated with batteries under high and low material recovery rates in recycling compared to the baseline. The findings highlight that retired LFP batteries can achieve a negative net carbon footprint in recycling only when undergoing the HMR method at the highest assumed recycling rate. Furthermore, the results indicate that, except for the DCR method, all other recycling methods result in additional environmental impact at low recycling rates. It underscores the importance of selecting suitable recycling methods based on the distinct chemical compositions of the batteries, as well as the significance of ensuring effective recycling rates in order to achieve carbon reduction objectives.

3.2. Two-stage learning curve model

3.2.1. Bootstrap regression results for learning rates

Fig. 5 presents the bootstrap regression results of the learning rates for various LIB applications, namely electric vehicle (EV), electric bus (EB), energy-designed (ED), and power-designed (PD) stationary energy

storage systems. The M2C stage regression estimates the learning rate for battery cell manufacturing to be $26.77 \pm 2.43\%$, with mineral costs serving as a floor for the battery cell costs. The second stage fitting of EV pack and SESS assembling is split into four individual applications with battery cell costs as a floor. From battery cell to pack (i.e., C2P) manufacturing, the learning rates are $20.54 \pm 4.29\%$ for electric vehicles and $18.54 \pm 10.58\%$ for electric buses. From battery cell to SESS (i.e., C2S) manufacturing, the learning rates are $18.49 \pm 8.44\%$ and $24.06 \pm 8.33\%$ for energy-designed and power-designed storage systems, respectively. Despite the broad uncertainty in historical SESS battery price data, there is a consistent downward trend over time. Moreover, compared to the second-stage learning, the first-stage (M2C) demonstrates a more consistent regression result with the historical data, leading to a narrower uncertainty interval. This finding can be attributed to the fact that the learning driver for battery cell cost reduction encompasses all chemistry types of LIB production volumes, and the learning effect has been relatively well-established due to its similarities with long-standing cell manufacturing for portable electronic devices. Additionally, the outcomes of this learning curve model lead to a more robust learning effect for battery prices compared to recent studies based on learning curves. The learning rates obtained in this study are approximately 4.05%–4.54% higher for EV batteries (Nykqvist et al., 2019; Hsieh et al., 2019; Schmidt et al., 2017) and 1.5–2 times of the previous study for SESS batteries (Kittner et al., 2017; Schmidt et al., 2019). These increased learning effects suggest that the reduction in battery production or assembly costs has outpaced the overall decrease in battery prices over the past decades. Notably, this analysis reveals a significantly higher learning rate for SESS batteries, which can be attributed to factors such as differences in the dataset and spillover effects resulting from the accelerated advancement and widespread use of EV batteries. These factors have contributed to a more rapid decline in SESS assembly costs.

3.2.2. Future battery price projections

Fig. 6 presents the projected global battery prices for various battery applications until 2030, obtained by calculating the weighted average using the production share by regions and market share by chemistry. The base case projection assumes that the costs of critical raw materials will continue to increase at their ten-year CAGR. The battery price projection curves demonstrate a gradually decelerating downward trend, especially for battery cells (represented by the gray lines). This

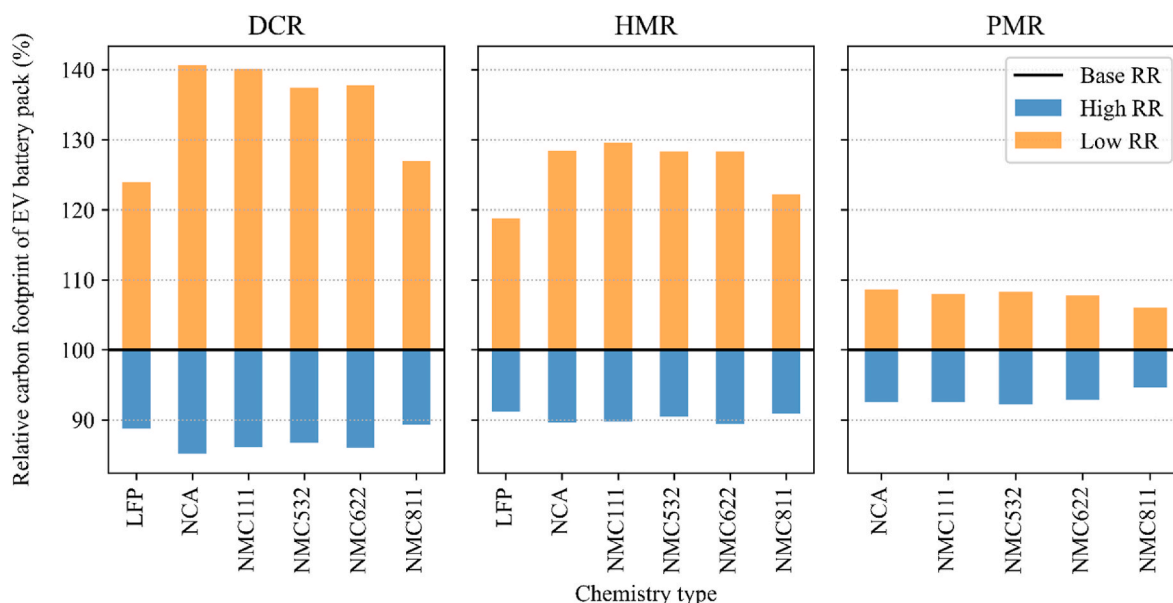


Fig. 4. Carbon footprint of EV battery pack on global average under high and low material recovery rates (RR) in recycling relative to the baseline in 2020.

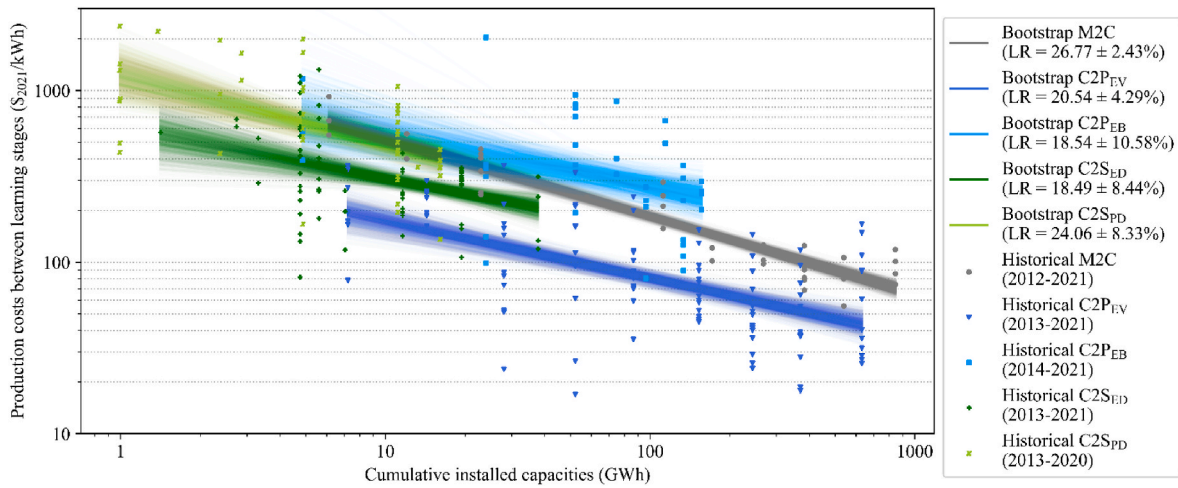


Fig. 5. Bootstrap regressions results of the two-stage learning curve model: the first stage is manufacturing from material to cell (M2C); the second stage is manufacturing either from cell to pack (C2P) for electric mobility or system (C2S) for stationary energy storage systems.

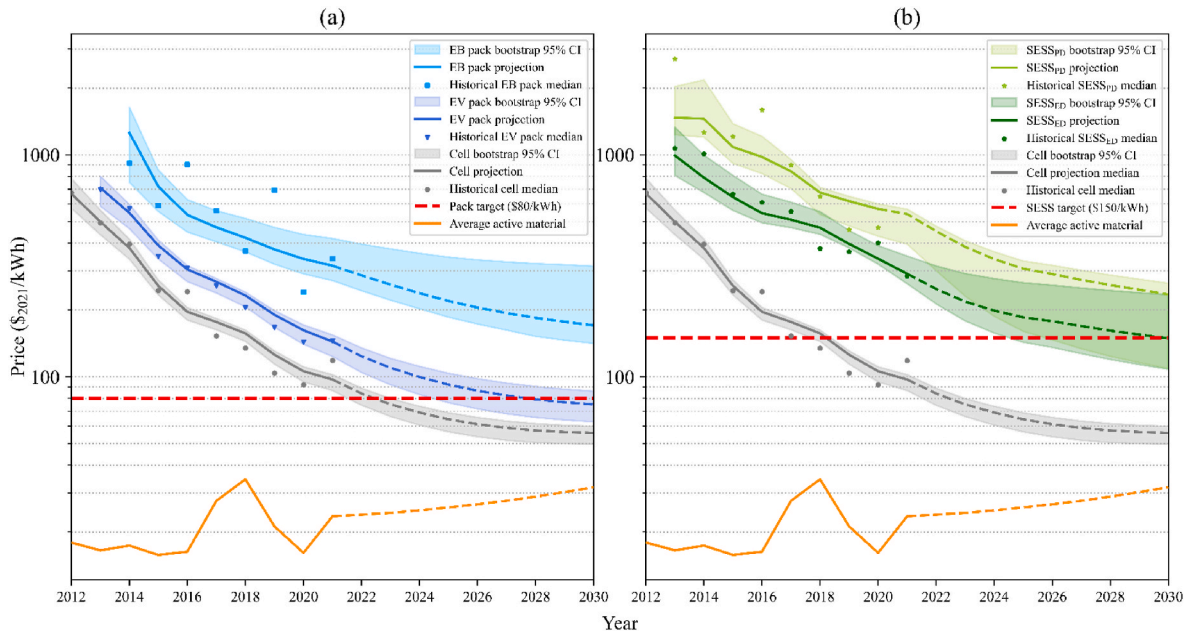


Fig. 6. Market average past and projected price trajectories of LIBs used for (a) electric vehicles and electric buses; (b) energy-designed and power-designed stationary energy storage systems.

trend is mainly attributed to the expected increase in mineral costs, which offset the cost reductions achieved through the learning effects of the cell manufacturing process. Consequently, the pace of battery price decline is projected to slow down, leading to an estimated average price in 2030 of approximately \$55.9/kWh (95% CI: \$49.8-\$60.0/kWh) for battery cells, \$75.1/kWh (95% CI: \$62.7-\$86.3/kWh) for battery packs in electric vehicles, \$170.7/kWh (95% CI: \$141.4-\$315.5/kWh) for battery packs in electric buses, and \$149.1/kWh (95% CI: \$108.2-\$234.4/kWh) for energy-designed and \$235.1/kWh (95% CI: \$109.1-\$264.4/kWh) for power-designed SESS at the system level by 2030. It is worth noting that despite having the same cell chemistry, the prices of battery packs for electric buses and power-designed SESS are generally higher compared to those for electric vehicles and energy-designed systems. This can be attributed to observed higher values of original data points, which may be influenced by differences in design requirements such as horsepower and power-to-energy ratios (Tsiropoulos and Tarvydas, 2018; California Environmental Protection Agency, 2016), as well as market mechanisms. Furthermore, the model results

suggest that projecting battery prices for electric vehicles involves less inherent uncertainty, indicated by narrower confidence intervals, compared to electric buses and stationary energy storage systems. It is primarily due to the larger sample size ($N = 117$), resulting in lower variability in the bootstrap distribution.

Given the inherent characteristics of commodities, mineral prices are known to be volatile and subject to uncertainties. Considering various price fluctuations and surges observed in recent years due to political events, the ten-year compound annual growth rate is selected as the base case scenario, which provides more reasonable (though potentially more optimistic) projections for battery prices. However, to account for future mineral price deviations, two additional scenarios are considered: a high growth rate scenario with a five-year CAGR and a zero-growth rate scenario assuming no change in mineral prices until 2030 (as defined in Table S14). Based on different mineral price growth scenarios (Fig. S7 and Fig. S8), the model predicts that the global weighted averages of LIB pack prices for electric vehicles will range from \$66.9/kWh to \$88.5/kWh in 2030. It indicates that fluctuations in critical material prices can

lead to a price differential of up to \$21.6/kWh on the final price, which cannot be mitigated through the learning effects of the cell manufacturing process. Moreover, the results also indicate that higher growth in mineral prices would make it unlikely for the average prices of EV battery packs to reach the target prices of \$80/kWh by 2030.

The impact of changing raw material prices on the final per-kWh prices varies across different cathode active materials. Fig. 7 illustrates the battery pack price trajectories of electric vehicles for each chemistry type under different mineral price growth scenarios (the price trajectory results of battery cells and other battery applications can be found in Figs. S9–S12). At the individual chemistry level, it is observed that increasing mineral costs will amplify the battery price differences between LFP and other nickel/cobalt-based LIBs on a per-kWh basis. The results indicate that LFP is the only cathode active material capable of achieving the price target of \$80/kWh by 2030, even under the high CAGR scenario. In contrast, for nickel/cobalt-based LIBs such as NCA and NMC, the price target can only be attained under the zero mineral price growth scenario. However, it is unlikely that mineral prices will remain fixed throughout 2030 (Donnelly, 2022), considering significant price increases for lithium, cobalt, and nickel, with respective rises of over 1030% (Economics, 2022b), 57% (Trading Economics, 2022c), and 112% (Trading Economics, 2022c) over the past two years (until December 2022). Furthermore, the results show that under the high CAGR scenario, the prices of nickel/cobalt-based batteries would rebound after 2027 because the continuously rising mineral prices will eventually outweigh the cost benefits derived from the learning effects of both battery cell and pack manufacturing processes.

3.3. The impacts of carbon pricing on battery price

The carbon price, functioning as a regulatory tool to internalize external carbon costs and reduce industrial emissions, serves as a crucial link between two subtopics explored in this study: life cycle assessment and techno-economic analysis. In order to understand the effects of carbon pricing, the market average prices of LIBs considering the carbon price imposed on the life cycle GHG emissions are analyzed. Adopting a life cycle perspective provides a comprehensive assessment of the expenses associated with the battery's carbon footprint throughout the supply chain—from raw material extraction and manufacturing to the EoL phase. Fig. 8 illustrates the impacts of carbon pricing on the projected average prices of EV battery packs (Figs. S13–S15 for other applications). The solid lines and dashed lines represent battery prices with carbon costs based on the highest and lowest carbon footprints derived from the life-cycle carbon footprint assessment shown in Fig. 2. Several noteworthy findings are as follows:

Firstly, recycling LIBs amplifies the benefits as the carbon price increases. The impact of the carbon price on battery pack prices is

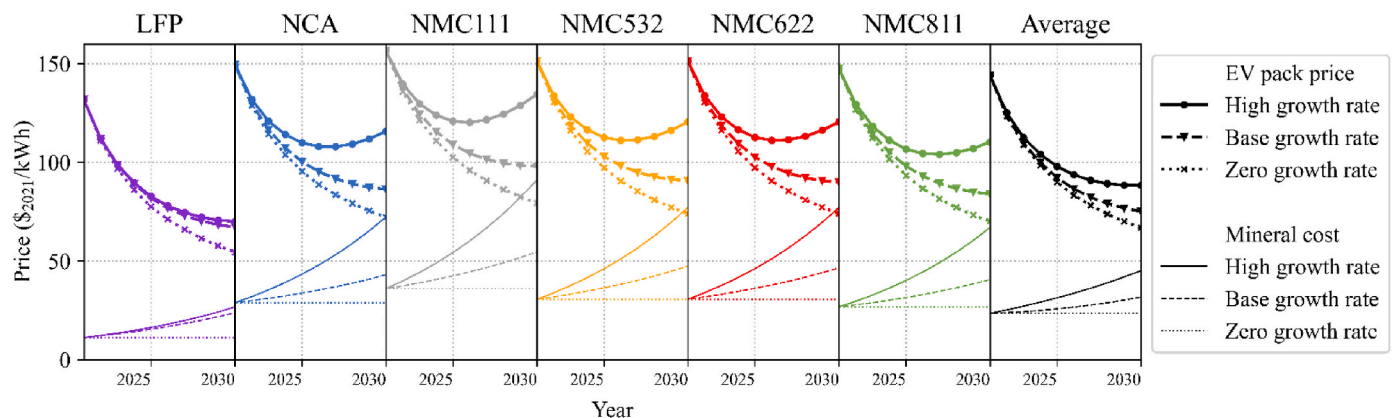


Fig. 7. Projected battery pack price for electric vehicles by different cathode chemistry types from 2021 to 2030, under the high (upper bound), base (middle), and zero growth rate (lower bound) scenarios for mineral costs.

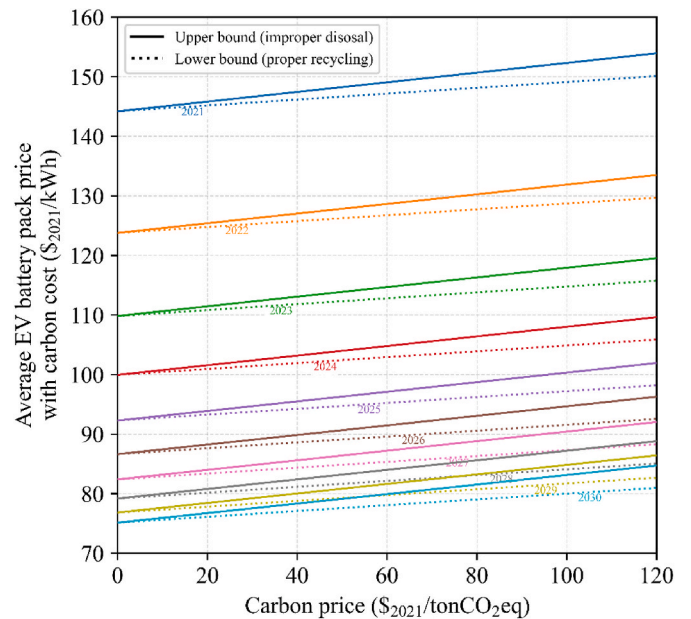


Fig. 8. Average LIB pack prices of electric vehicles from 2021 to 2030 considering different carbon prices for improper disposal (incineration) and proper recycling (DCR) EoL treatments.

approximately 64% higher for LIBs disposed of through incineration compared to those recycled using the DCR method. It highlights the increasing importance of recycling in a future where more countries or economies adopt carbon pricing policies. Secondly, the slopes of improper disposal and proper recycling from 2020 to 2030 are nearly identical (with only a 1.1–1.6% difference), suggesting that the effects of declining grid carbon intensity in the next decade on LIB emissions mitigation are relatively negligible compared to the influence of a rising carbon price. Thirdly, carbon pricing significantly offsets the cost savings achieved through learning effects in battery pack manufacturing. In 2030, the projected EV battery pack prices are expected to return to the target of \$80/kWh when the carbon price reaches \$100 per tonCO₂eq (with proper recycling) and \$61 per tonCO₂eq (with improper disposal). According to the International Monetary Fund (IMF), a global average carbon price of at least \$75 by the end of 2030 is necessary to achieve climate goals (Parry, 2022). This underscores the importance of battery recycling not only for climate change mitigation but also for maintaining battery economic competitiveness.

3.4. Limitations and future outlook

This study primarily addresses emerging carbon-related concerns and offers an environmentally and economically oriented assessment of LIBs in response to CBAM and global climate change imperatives. However, in addition to GWP, there are other indicators that can be evaluated for conducting LCA, such as water resource consumption, metal resource scarcity, etc. Modeling all environmental impacts in detail and monetizing them appropriately for different regions, time scales, and policies will be an important issue in the future. Future research endeavors should delve into these parameters, providing a more holistic perspective. Furthermore, this study recognizes the potential volatility of input variables within the learning curve model, particularly the costs of critical raw materials, which can fluctuate due to market dynamics or international geopolitical tensions. The two-stage learning curve model proposed in this study can effectively address this situation, enabling subsequent researchers to generate timely forecasts based on real-time prices of critical raw materials at the point of prediction. Lastly, in the spirit of fostering collaboration within the academic community, the data used in this study have been made available in supplementary files ([Supplementary data](#) (which is an Excel file) and [Supplementary information](#) (which is a Word file)). This resource is intended to facilitate and inspire subsequent studies, enabling a more robust and collective exploration of LIBs from various angles and dimensions.

4. Conclusion

This study provides a thorough evaluation of LIBs, encompassing both their carbon footprint and economic dimensions. This assessment is achieved by combining an LCA approach to measure carbon footprint and TEA. The carbon footprint assessment results unveil that the carbon footprint of an EV battery pack ranges from 49.2 to 81.1 kgCO₂eq/kWh, contingent on the selected end-of-life treatment. Notably, NCA and NMC622 battery chemistries demonstrate lower carbon emissions compared to others. Regionally, LIBs manufactured in China exhibit the highest life cycle carbon footprint, while Europe records the lowest. Incorporating recycling into the battery life cycle holds substantial potential for carbon footprint reduction, with the potential to lower it by 28%–46% when compared to incineration or landfill, depending on the battery chemistry and recycling methods.

The future price projections for LIBs are inherently uncertain and influenced by various factors including learning curve effects in manufacturing, critical mineral prices, and external carbon costs. This study introduces a two-stage learning curve model that considers material costs and learning rate regression, driven by cumulative battery installation capacities. The findings indicate a projected price of \$75.1/kWh (95% CI: \$62.7–\$86.3/kWh) on average for battery packs in electric passenger vehicles by 2030. However, only the LFP battery for EVs showed potential to reach the target price of \$80/kWh by 2030, even with a high compound annual growth rate. Nonetheless, it's crucial to note that the price decline due to learning effects is anticipated to be counterbalanced by carbon regulations when factoring in carbon costs on LIBs. Effective recycling practices can play a pivotal role in acquiring carbon footprint credits from secondary materials, consequently reducing life cycle emissions-related costs within the battery supply chain.

As the global average carbon price is projected to reach \$75 per ton of carbon dioxide by 2030 in line with climate objectives, EV battery pack prices are expected to rise to \$78.8/kWh (with direct cathode recycling) to \$81.1/kWh (with incineration). This underscores the imperative of prioritizing effective recycling practices to uphold the economic and environmental competitiveness of batteries in the context of climate change, the expanding use of e-mobility, and the growing adoption of renewable energy sources.

CRedit authorship contribution statement

Wei-Hsuan Chen: Formal analysis, Investigation, Writing – original draft, Visualization. **I-Yun Lisa Hsieh:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2023.139045>.

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